Prem Rathod | Data Scientist | Machine Learning Engineer

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Professional Summary

Results-driven Final Year M.Sc. IT student specializing in **Data Management and Visual Analytics**. Proficient in building end-to-end Machine Learning pipelines, from data wrangling in SQL and Python to deploying Deep Learning models using TensorFlow and OpenCV. Awarded **First Class with Distinction**, with a strong focus on Computer Vision and Predictive Modeling. Seeking to leverage technical expertise to solve complex real-world data challenges.

Technical Skills

Languages: **Python** (Pandas, NumPy, Scikit-learn), **R**, **SQL** (Advanced Joins, CTEs, Optimization)

AI & Machine Learning: **Computer Vision** (OpenCV), **Deep Learning** (TensorFlow, Keras, CNN, DenseNet)

Data Engineering: MS Excel (Power Query, Pivot Tables), ETL Processes, **Snowflake**, AWS, MySQL, SQLite

Visualization: **Power BI**, Tableau, Matplotlib, Seaborn

Web Frameworks: Flask (for ML Model Deployment), Tkinter

Education

Gujarat University **Ahmedabad**

Integrated M.Sc. in Information Technology, Expected May 2026 *2021–2026*

- **Academic Standing:** First Class with Distinction (Semesters 1-9).
- **Relevant Coursework:** Machine Learning, AI-driven Applications, Advanced Data Management, Visual Analytics.

Technical Projects

Python, TensorFlow, DenseNet

Clothing Recommendation System *Deep Learning*

- Developed a visual search engine using **DenseNet** for high-level feature extraction from fashion images.
- Calculated **Cosine Similarity** metrics to provide high-accuracy product recommendations based on visual style.
- Preprocessed and augmented large-scale datasets to improve model generalization and recommendation relevance.

OpenCV, SQLite, Flask

Face Recognition Attendance System *Computer Vision*

- Engineered a real-time face recognition pipeline utilizing **Deep Learning** models for accurate employee/student identification.
- Designed a secure backend with **SQLite** to log timestamps and integrated a **Flask** web dashboard for administrative monitoring.
- Optimized the recognition algorithm for low-latency performance on standard webcam hardware.

SQL, Power BI, Python

Sales Performance Intelligence Dashboard *Data Analytics*

- Performed complex ETL tasks using **Python (Pandas)** and **Power Query** to unify data from multiple business schemas.
- Wrote optimized **SQL CTEs** to track regional performance KPIs and identify sales bottlenecks.
- Created interactive dashboards that pinpointed a **15% drop** in specific regional segments, facilitating strategic decision-making.

Certifications & Interests

Interests: Predictive Modeling, Neural Network Optimization, Competitive Programming, Data Storytelling.

Certifications: Advanced Data Analytics with Python; SQL for Data Science (Specialization).